

RHCSA RHEL 9 — Premium Exam Q&A

Last Updated: May 2023 | Node 1 + Node 2

NODE 1

Q1 — Set hostname and Assign IP address

hostname = node1.lab.network17.com

ip address = 172.25.250.11/24

gateway = 172.25.250.254

dns = 172.25.250.254

Also enable PermitRootLogin in sshd_config.

Solution:

```
# hostnamectl set-hostname node1.lab.network17.com
# hostname
# nmcli connection
# nmcli connection modify "Wired connection 1" ipv4.addresses 172.25.250.11/24
# nmcli connection modify "Wired connection 1" ipv4.gateway 172.25.250.254
# nmcli connection modify "Wired connection 1" ipv4.dns 172.25.250.254
# nmcli connection modify "Wired connection 1" ipv4.method static
# nmcli connection up "Wired connection 1"
# ping 172.25.250.11
# ping 172.25.250.254
# vim /etc/ssh/sshd_config
Find line: #PermitRootLogin prohibit-password
Remove # and change to: PermitRootLogin yes
:wq!
# ssh root@node1
```

Q2 — Create a YUM/DNF repository

AppStream baseurl:

http://classroom.example.com/content/rhel8.0/x86_64/dvd/AppStream

BaseOS baseurl: http://classroom.example.com/content/rhel8.0/x86_64/dvd/BaseOS

Solution:

```
# vim /etc/yum.repos.d/app.repo
[AppStream]
name=app
baseurl=http://classroom.example.com/content/rhel8.0/x86_64/dvd/AppStream
gpgcheck=0
enabled=1
[BaseOS]
name=base
```

```
baseurl=http://classroom.example.com/content/rhel8.0/x86_64/dvd/BaseOS
gpgcheck=0
enabled=1
:wq
# yum clean all
# yum repolist all
```

Q3 — Configure web server on non-standard port 82

- Web server must serve content from /var/www/html
- Port 82 must be open in SELinux and firewall
- Service must start automatically at boot

Solution:

```
# semanage port -l | grep http
# semanage port -a -t http_port_t -p tcp 82
# semanage port -l | grep http
# firewall-cmd --permanent --add-port=82/tcp
# firewall-cmd --reload
# firewall-cmd --list-ports
# yum install httpd -y
# systemctl start httpd.service
# systemctl enable httpd.service
# vim /etc/httpd/conf/httpd.conf
(search for /Listen and verify port)
Add:
<VirtualHost 172.25.250.11:82>
ServerName serverb.lab.example.com
DocumentRoot /var/www/html
</VirtualHost>
:wq
# httpd -t
# systemctl restart httpd.service
Verify:
# curl http://172.25.250.11:82/file1
```

Q4 — Users and groups management

- Create group: admin
- Users: harry and natasha (supplementary group: admin)
- User: sarah (nologin shell)
- Password for all users: given in exam

Solution:

```
# groupadd admin
# useradd -G admin harry
```

```
# useradd -G admin natasha
# useradd -s /sbin/nologin sarah
# passwd --stdin harry
# passwd --stdin natasha
# passwd --stdin sarah
Verify:
# id harry
# id natasha
# su - sarah (should not be available)
# tail /etc/passwd
# tail /etc/group
```

Q5 — Create a directory and assign permissions (collaborative directory)

- Directory: /common/admin
- Group owner: admin
- Permissions: rwx for owner and group, none for others
- SGID: new files inherit group ownership

Solution:

```
# mkdir -p /common/admin
# ls -ld /common/admin
# chgrp admin /common/admin
# chmod 770 /common/admin
# chmod g+s /common/admin
# ls -ld /common/admin
Verify:
# su - harry
# touch /common/admin/file1
# ls -l /common/admin/file1
```

Q6 — Configure AutoFS

- User: production5
- Remote share: servera.lab.example.com:/home-directories/production5
- Local mount point: /localhome/production5
- Mount automatically on login

Solution:

```
# yum install autofs -y
# systemctl start autofs.service
# systemctl enable autofs.service
# getent passwd production5
# vim /etc/auto.master
Add last line (do not edit existing contents):
/localhome /etc/auto.misc
```

```
:wq
# vim /etc/auto.misc
Add last line:
production5 -rw,soft,intr servera.lab.example.com:/home-directories/production5
:wq
# systemctl restart autofs
Verify:
# su - production5
# pwd
# df -hT
# exit
```

Q7 — Configure crontab for user harry

Examples from PDF (actual times/commands will differ in exam):

- At 12:30 pm: /bin/echo "hello"
- Every 3 minutes: /bin/echo "hello"
- Every 2 minutes: logger user.debug "hello"

Solution:

```
# crontab -eu harry
Example entries (use values given in your exam):
30 12 * * * /bin/echo "hello"
*/3 * * * * /bin/echo "hello"
*/2 * * * * logger user.debug "hello"
Verify:
# crontab -lu harry
```

Q8 — NTP configuration

- Configure chrony to use: classroom.example.com as NTP server

Solution:

```
# vim /etc/chrony.conf
Comment out existing server/pool iburst line(s)
Add new line:
server classroom.example.com iburst
:wq
# systemctl restart chronyd.service
Verify:
# chronyc sources -v
```

Q9 — Find files owned by a specific user and copy them

- Find all files owned by user: sarah
- Copy them to: /root/find.user

Solution:

```
# mkdir /root/find.user
# find / -user sarah -type f
# find / -user sarah -type f -exec cp {} /root/find.user \;
Verify:
# ls -a /root/find.user
```

Q10 — Grep the string "home" from /etc/passwd and redirect output

- Search for "home" in /etc/passwd
- Save result to: /root/search.txt

Solution:

```
# grep home /etc/passwd
# grep home /etc/passwd > /root/search.txt
Verify:
# cat /root/search.txt
```

Q11 — Create a user with specific UID and set password

- Username: alies
- UID: 1326
- Password: redhat (given in exam)

Solution:

```
# useradd -u 1326 alies
# passwd --stdin alies
redhat
Verify:
# id alies
```

Q12 — Backup a directory using tar with compression

- Create archive of: /var/tmp
- Output: /root/test.tar.gz (gzip) or /root/test.tar.bz2 (bzip2)

Solution:

```
For gzip (.gz):
# tar -zcvf /root/test.tar.gz /var/tmp
For bzip2 (.bz2):
# tar -jcvf /root/test.tar.bz2 /var/tmp
```

Q13 — Container Question 1 — Build a container image named 'monitor'

- Login with user walhalla on node1
- Download Containerfile using wget (URL given in exam)
- Build image named: monitor
- Login to registry (credentials given in exam)

Solution:

```
# ssh walhalla@node1
$ wget <given URL for Containerfile>
$ ls
$ podman login <given registry link>
Username: admin
Password: redhat321
$ podman build -t monitor .
$ podman images
```

Q14 — Container Question 2 — Run container as systemd service

- Container name: ascii2pdf
- Image: monitor (built in Q13)
- Volume: /opt/files -> /opt/incoming:Z
- Volume: /opt/processed -> /opt/outcoming:Z
- Create systemd user service: container-ascii2pdf
- Service must survive reboot

Solution:

```
As root (Tab 1):
# mkdir /opt/files /opt/processed
# chmod 777 /opt/files /opt/processed
# chown student:student /opt/files /opt/processed
As user walhalla (Tab 2):
$ podman run -d --name ascii2pdf -v /opt/files:/opt/incoming:Z -v /opt/processed:/opt/outcoming:Z localhost/monitor:latest
$ podman ps
$ mkdir -p ~/.config/systemd/user
$ cd ~/.config/systemd/user
$ podman generate systemd --name ascii2pdf --files --new
$ podman stop ascii2pdf
$ podman rm ascii2pdf
$ podman ps
$ systemctl --user daemon-reload
$ systemctl --user start container-ascii2pdf.service
$ systemctl --user enable container-ascii2pdf.service
$ podman ps
As root:
# loginctl enable-linger walhalla
# reboot
Verify after reboot:
$ podman ps
```

Q16.1 — Set default file/directory permissions using umask for user natasha

- Configure umask so new files get permission: r----- (400)
- New directories get permission: r-x----- (500)
- Must be persistent (set in .bash_profile)

Solution:

```
# su - natasha
# ls -a (find .bash_profile)
# vim .bash_profile
Add last line (do not disturb existing contents):
umask 277
:wq!
# source .bash_profile
Verify:
# mkdir test
# ls -ld test
# touch testfile
# ls -l testfile
```

Q16.2 — Set password expiry for all users

- Max password age: 20 days (value given in exam)

Solution:

```
# vim /etc/login.defs
Find line: PASS_MAX_DAYS 99999
Change to: PASS_MAX_DAYS 20
:wq!
```

Q16.3 — Assign sudo privilege to group admin without password

Solution:

```
# vim /etc/sudoers
:se nu (find line ~111)
Add:
%admin ALL=(ALL) NOPASSWD: ALL
```

Q16.4 — Configure application environment variable for user alies

- Variable name: RHCSA
- Value: "Welcome to Advantage Pro"
- Must display on login

Solution:

```
# su - alies
# ls -a
# vim .bash_profile
```

```
Add at last line:
RHCSA="Welcome to Advantage Pro"
export RHCSA
echo $RHCSA
:wq!
# source .bash_profile
# logout
# su - alies (verify message is shown)
```

Q16.5 — Create a shell script that finds and copies files

- Script name: mysearch
- Find files under /usr/share/ with size < 1M
- Copy found files to /root/myfiles/
- Script variations depend on exam (size, permissions)

Solution:

```
# mkdir /root/myfiles
# vim mysearch
#!/bin/sh
find /usr/share/ -type f -size -1M -exec cp {} /root/myfiles/ \;
(Alternative – with setuid bit u+s):
find /usr/share/ -type f -size -1M -perm /4000 -exec cp {} /root/myfiles/ \;
(Alternative – with setgid bit g+s, size range):
find /usr/share/ -type f -size +800M -size -900M -perm /2000 -exec cp {}
/root/myfiles/ \;
:wq!
# chmod +x mysearch
# ./mysearch
Verify:
# ls -a /root/myfiles/
```

NODE 2

QN2-1 — Break the root password on Node 2 (password reset via rd.break)

Solution:

1. Open console and reboot the machine
 2. In GRUB loader select the rescue/kernel entry
 3. Press 'e' to edit the boot entry
 4. Find the line starting with 'linux', press End key
 5. Append: rd.break at the end of the line
 6. Press Ctrl+x to boot
- In the emergency shell:
- ```
mount -o remount,rw /sysroot
```

```
chroot /sysroot
passwd --stdin root
northate (use password given in exam)
touch /.autorelabel
exit
exit
(system reboots automatically)
After reboot:
vim /etc/ssh/sshd_config
Find: #PermitRootLogin prohibit-password
Change to: PermitRootLogin yes
:wq!
ssh root@node2
```

## QN2-2 — Configure YUM/DNF repository (same as Node 1 Q2, different node)

### Solution:

```
vi /etc/yum.repos.d/app.repo
[AppStream]
name=app
baseurl=http://classroom.example.com/content/rhel8.0/x86_64/dvd/AppStream
enabled=1
gpgcheck=0
[BaseOS]
name=base
baseurl=http://classroom.example.com/content/rhel8.0/x86_64/dvd/BaseOS
enabled=1
gpgcheck=0
:wq
yum clean all
yum repolist
yum install vim -y
```

## QN2-3 — Create a swap partition of 512MB and mount it permanently

- Device: /dev/vdb
- Size: 512M
- Type: Linux swap (82)
- Must be mounted permanently via /etc/fstab

### Solution:

```
free -m (check existing swap)
lsblk
fdisk /dev/vdb
Command: n
```

```

Select: p (primary)
Partition number: 3
First sector: Enter (default)
Last sector: +512M
Command: t
Partition number: 3
Hex code: 82 (Linux swap)
Command: w
udevadm settle
lsblk
mkswap /dev/vdb3
vim /etc/fstab
Add last line (do not change existing contents):
/dev/vdb3 swap swap defaults 0 0
:wq
swapon -a
Verify:
free -m

```

## QN2-4 — Create an LVM partition, format and mount permanently

- Device: /dev/vdb
- Partition size: 2G, type: LVM (8e)
- VG name: datastore, PE size: 8M
- LV name: database, size: 50 extents
- Filesystem: ext3
- Mount point: /mnt/database

### Solution:

```

lsblk
fdisk /dev/vdb
Command: n
Select: p (primary)
Partition number: 2
First sector: Enter (default)
Last sector: +2G
Command: t
Partition: 2
Hex code: 8e (Linux LVM)
Command: w
udevadm settle
lsblk
vgcreate -s 8M datastore /dev/vdb2
vgdisplay datastore
lvcreate -l 50 -n database datastore

```

```
lvdisplay /dev/datastore/database
mkfs.ext3 /dev/datastore/database
mkdir /mnt/database
vim /etc/fstab
Add last line (do not disturb existing contents):
/dev/datastore/database /mnt/database ext3 defaults 0 0
:wq
mount -a
Verify:
df -hT
lsblk
```

## QN2-5 — Resize (extend) the LVM partition

- Extend LV `/dev/mapper/datastore-database`
- New size: 100 extents (or 100M depending on exam)
- Filesystem must resize online

### Solution:

```
lsblk
df -hT
lvextend -l 100 -r /dev/mapper/datastore-database
(or, if size in MB):
lvextend -L 100M -r /dev/mapper/datastore-database
Verify:
lsblk
df -hT
```

## QN2-7 — Set the recommended tuned profile as default

### Solution:

```
yum install tuned -y
systemctl start tuned.service
systemctl enable tuned.service
tuned-adm active (check current active profile)
tuned-adm recommend
(note the recommended profile, e.g. virtual-guest)
tuned-adm profile virtual-guest
Verify:
tuned-adm active
```